Abstract

Intro

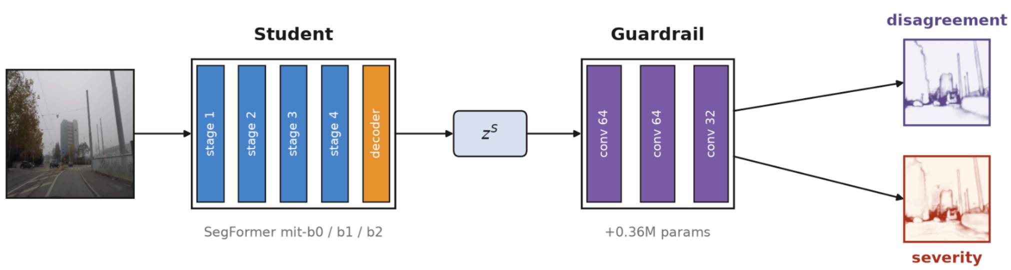
Problem Statement

Related Work

The Guardrail

We propose…

We propose a light guardrail operating on frozen student logits, predicting dense per-pixel student-teacher disagreement and severity. A strong teacher model T (SegFormer-B5) is first used to distill the student S. Both networks are then frozen, and we train the guardrail G on top of the student's logits to predict the severity and student-teacher disagreement. We isolate the effect of the supervision signal by comparing the two cases of supervision by: (a) teacher-derived dense targets, and (b) ground-truth-derived dense targets, while fixing the guardrail architecture, optimizer, and corruption augmentation.



Caption: Overview of the guardrail pipeline. The guardrail reads the student per-pixel logits (C=19 channels) concatenated with the student's decoder feature map (F; channels=256 for mit-b0, 512 for b1/b2), and outputs a dense per-pixel map of severity and student-teacher disagreement.

Notation

Notation section with in-line LaTeX equations.
Not sure if this is correct, use the correct latex math
For an input image x, let $z^S(x), z^T(x) \in \mathbb{R}^{H\times W\times C}$ denote the student and teacher logit tensors, where $H \times W$ is the output spatial resolution, and C is the number of semantic classes. Let $\hat{y}^S_{ij} = \arg\max_c z^S_{ijc}$, $\hat{y}^T_{ij} = \arg\max_c z^T_{ijc}$ be the student and teacher predictions at pixel (i, j), let y_{ij} denote the ground-truth label, and let $m_{ij} \in \{0, 1\}$ be a valid-pixel mask that excludes ignored labels. We write $\mathrm{CE}(z, y) = -\log(\mathrm{softmax}(z)_y)$ for pixelwise cross-entropy.
Add another sentence or two with in-line equations for the metrics used, including that risk= 1 - miou

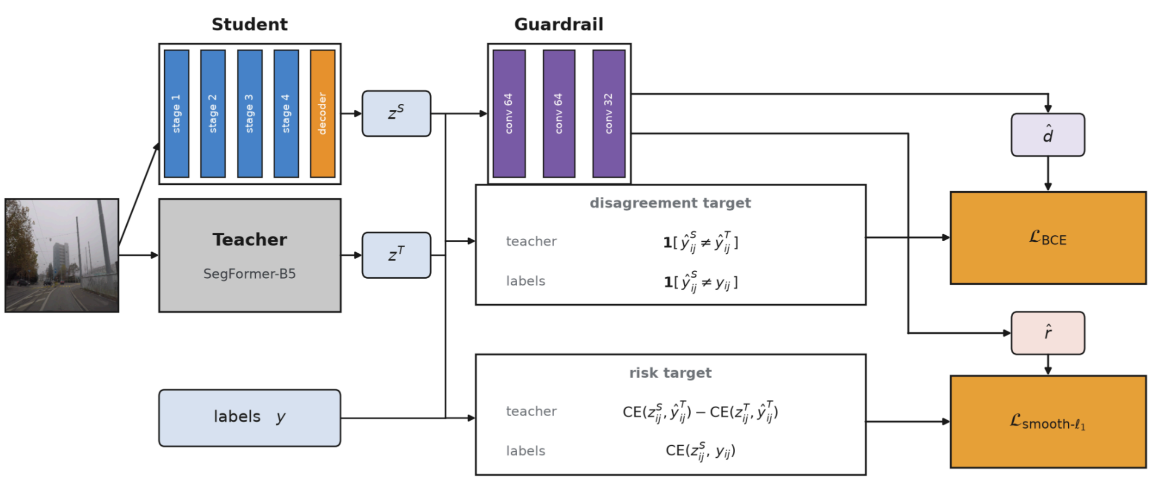
Guardrail Architecture

G is a light convolutional head that reads the frozen student logits (C=19 channels) concatenated with the student's decoder feature map (F; channels=256 for mit-b0, 512 for b1/b2) bilinearly resized to the logit resolution. Both inputs are detached, and no gradients flow into S. A shared encoder of three 3x3 convolutions, followed by batch normalization and ReLU, with channel dimensions (19+F) → 64 → 64 → 32 feeds two 1x1 convolutional output heads. The first output head predicts the dense disagreement map $d \in [0, 1]^{H\times W}$, and the second predicts the continuous severity map $\hat{s} \in \mathbb{R}^{H\times W}$. The head is small (0.21M parameters for b0, 0.36 for b1/b2. So 2-6% of student parameters), and does not call the teacher at inference. In the multi-task configuration, both outputs share an encoder and are optimized jointly. Full layer shapes and hyperparameters are given in Appendix D.

Supervision

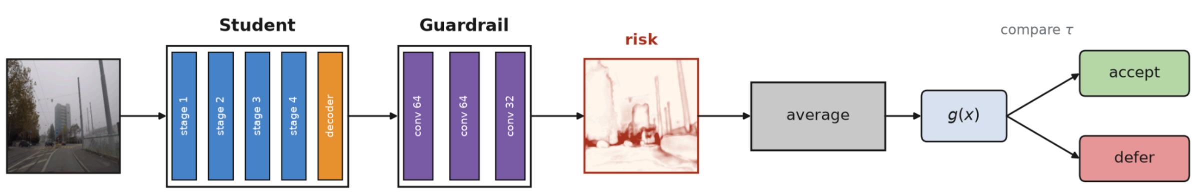
We compare two supervision cases for training G, which only differ in how the dense heads are constructed. In both cases, the guardrail receives the student logits z^2 and outputs the pair $(\hat{d}, \hat{r})$. In the teacher supervised case, the binary target indicates student-teacher disagreement at each pixel $d^T_{ij} = 1[\hat{y}^S_{ij} \neq \hat{y}^T_{ij}]$. The continuous target (severity) is the unclipped, signed difference between the student and teacher's cross-entropy at the teacher's predicted class **Not sure if this is correct, use the correct math**$r^T_{ij} = \mathrm{CE}(z^S_{ij}, \hat{y}^T_{ij}) - \mathrm{CE}(z^T_{ij}, \hat{y}^T_{ij}) = \log \frac{p^T_{ij}(\hat{y}^T_{ij})}{p^S_{ij}(\hat{y}^T_{ij})}$. In the ground truth supervision case, the binary targets indicate whether the student's prediction is incorrect compered to the labeld class: $d^{GT}_{ij} = 1[\hat{y}^S_{ij} \neq y_{ij}]$, and the continuous severity target is the student's cross entropy with respect to the ground truth targets $r^{GT}_{ij} = \mathrm{CE}(z^S_{ij}, y_{ij})$. This allows for two training cases that isolate the source of supervision, to identify whether the teacher-derived dense severity targets transfer better under domain shift than dense targets derived from source-domain correctness.

Training

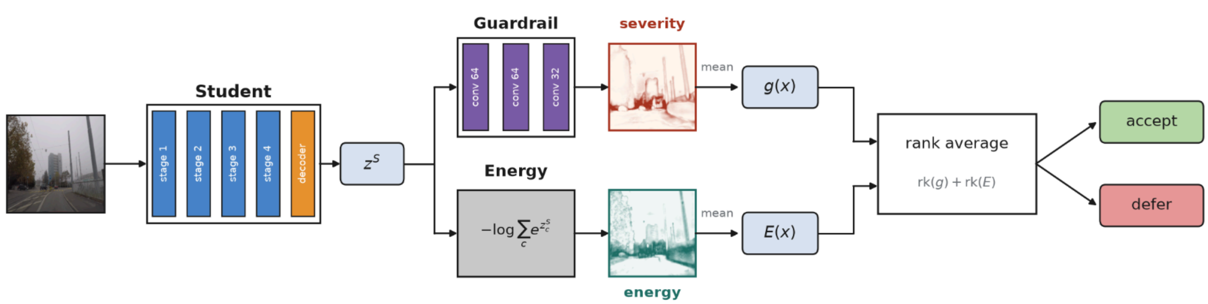


The teacher T is a SegFormer-B5 model (87.4M parameters) **CITE**, and the student S is a SegFormer b0/b1/b2 model. S is first pretrained on Cityscapes with standard semantic segmentation supervision. Next, S is distilled from the frozen teacher with a combination of segmentation loss, logit distillation with KL divergence (Temperature \ $\tau = 2$), and a pairwise-affinity feature loss inspired by structured distillation for dense prediction **Cite**. After this stage, all subsequent learning is on the guardrail head G and operates on detached student logits. The disagreement head is trained with masked binary cross-entropy loss, and the severity head is trained with masked smooth-ℓ1 loss, averaged over valid pixels. The full guardrail objective is LG = λdis LBCE(ˆd, d) + λrisk LSℓ1 (r, r ˆ), where (d, r) denotes either the teacher-supervised targets (d T , rT) or the ground-truth-supervised targets (d GT, rGT). Single-task variants are trained by setting one loss weight to zero. To broaden the supervision signal, we apply online corruption augmentation during guardrail training. Each training image is perturbed with probability $p=0.5$ by a sampled corruption including fog, blur, noise, and underexposure. Whenever a corruption

Inference



Guardrail + post-hoc



Post-hoc scores such as Energy and MaxLogit are fixed non-linear functions of the student's logits, so they carry no information beyond z^2. Although G already receives the logits and student features, a shallow convolution struggles to recover these global logit-magnitude statistics on it's own. We therefor introduce a variant that concatenates per-pixel energy and max-logit as two additional input channels to G. This widens the first layer's input, and the head, training procedure, and targets are otherwise unchanged.

Teacher signal under shift.

	mIoU		Disagreeing pixels (% of valid pixels)			
	Student	Teacher	Any	Teacher right	Student right	Both wrong
Cityscapes (in-domain)	.691	.774	4.2	2.7	1.1	0.4
ACDC (adverse conditions)	.405	.517	16.6	9.4	2.4	4.7
IDD (India)	.459	.532	8.7	4.7	2.1	1.9
BDD (US)	.445	.554	10.6	6.7	2.1	1.9

Caption: Breakdown of mIoU and the pixels where the student and teacher disagree (mit-b1). We see the dominant case for out-of-distribution is the teacher advantage which can be recovered with teacher labels for the guardrail.

It seems counterintuitive to train a reliability guardrail head on a teacher's predictions rather than the ground-truth labels, however, a ground-truth target considers all errors identical, but the teacher errors are often artifacts of the image that no abstention or deferral can cover. A teacher target is only worth training if the cases where the student errors while the teacher is correct are meaningful. Table \ref{} shows this case as the teacher is often right relative to the student is abundant and thus, there is room for student improvement to train the guardrail on. For in-domain cases, the student-teacher disagreement is low and the guardrail should then be trained on ground-truth labels, but for out of distribution cases, we see a significant gap of 4.7-9.4% of valid pixels, showing the teacher target has room to improve the guardrail. There is still the case where the student is correct but the teacher is wrong, and we see this is much less frequent than the teacher correct case, yet still must be considered. The severity head thus considers the ground-truth labels as it computes $CE(student, y) - CE(teacher, y)$ while the disagreement head ignores the labels.

Experiments

Datasets

We used Cityscapes (500 images) for the in-domain dataset because of X. ACDC (406 images) consists of fog, night, rain, and snow categories and fixes the geometry while varying the conditions, isolating adverse-condition shift. Additionally, IDD (978 images) and BDD (1000 images) instead vary the scene itself, including road layout, traffic composition, and geography, to test whether the guardrail generalizes to the overall problem.

Baselines

We compare against MSP, temperature-scaled MSP, MC-Dropout, MaxLogit, Entropy, Energy, and Deep Ensambles. Each baseline reduces the dense per-pixel score to an image-level score by averaging over the same valid pixels $g(x)$. MC-Dropout (4 passes) and the three member Deep Ensamble use predictive entropy of the mean softmax, and temperature scaling use a fixed $\tau=2$. Multi-task is the default unless stated otherwise. Our controlled comparison of teacher vs. ground-truth supervision uses the SegFormer-b1 student trained on three seeds (42/137/256) and averaged over them. Confidence-feature results (\ref) use five seeds across three backbones (b0/b1/b2).

Metrics

We consider per-image quality as mIoU: the mean IoU over the classes that appear in the union of the ground truth and predictions, computed over valid pixels only. We label an image a failure if $rho = 1 - mIoU$ is in the worst 20% of the images scored. This is a relative ranking rather than an absolute correctness ranking, and the cutoff is evaluated retrospectively rather than at a deployable threshold, and an equally poor image in a uniformly hard dataset may fall outside of this quintile. We therefore read out numbers as a failure ranking benchmark. For IDD and BDD, we read the 80th percentile cutoff as a single global value over the dataset, while for ACDC, it is computed separately within each condition and applied to the pooled set, so one uniformly difficult split does not dominate the failure set. F-AUROC ranks a score against this label over all images, and confident failure AUROC (CF-AUROC) restricts the ranking to the confident subset with image-level MSP \geq \tau where \tau is the mean over valid pixels of the per-pixel maximum softmax probability for \tau \in \{0.85, 0.95, 0.97\}. CF-AUROC finds among images the deployed student considers confident, which score best ranks the failures. Part of MSP's decline at large \tau is a consequence of conditioning on MSP. We only depart from the per-condition ACDC cutoff in \ref{}, because the purpose is to measure the between-condition signal that a per-condition cutoff removes. Under this global cutoff, night accounts for almost all failures.

Results

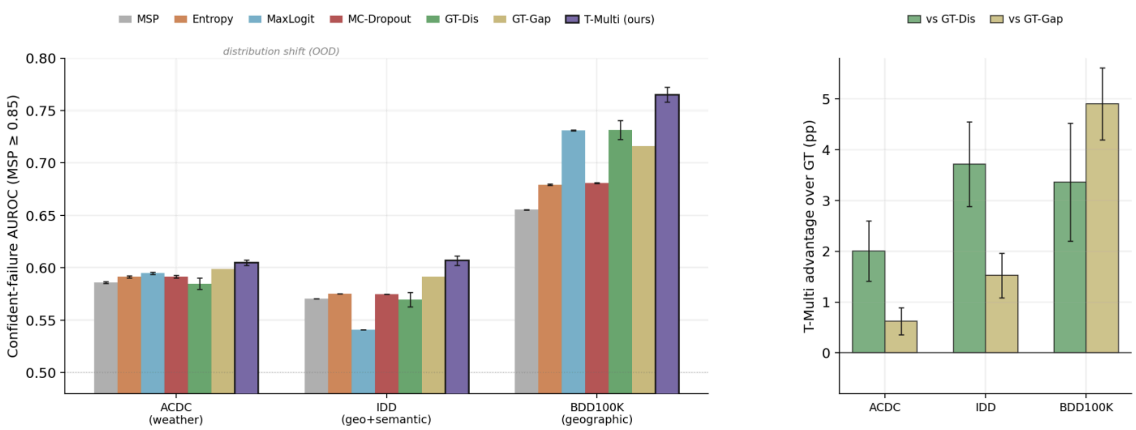
Comparison with Post-hoc methods

We compare with post-hoc metrics and this is the results for CF-AUROC @ \tau = 0.85.

	MSP	Ent	MaxL	Energy	MC-D	DeepEns	GT-Dis	GT-Gap	**T-Multi**	Δ_{GT}
ACDC	.587	.592	.596	.594	.593	**.605**	.587	.595	**.605**	**+1.0**
IDD	.573	.578	.543	.539	.577	.590	.570	.595	**.607**	**+1.2**
BDD	.657	.680	.732	.730	.682	.723	.731	.722	**.765**	**+3.5**

Caption: CF-AUROC @ MSP $\geq$ 0.85 (mit-b1), a near-full-coverage operating point, evaluated over three training seeds (42/137/256) with low per-seed variation (std $\leq$ .007 for every head and dataset).

Table \ref{} is the controlled experiment for the supervision signal, as T-Multi, GT-Gap, and GT-Dis share the same architecture, optimizer, and seeds, so Δ_{GT} attributes the difference from the target alone. T-Multi is the strongest score on all three datasets, matching a three-member Deep Ensamble on ACDC and beating it on IDD and BDD without training or deploying two extra students. The margins over the best ground truth head are small on ACDC and IDD but larger on BDD, though absolute values near 0.6 suggest that ranking confident failures under shift is far from solved. The table thus shows that for the same head and no teacher call at test-time, a teacher-relative dense target is a stronger learning signal than ground-truth correctness on out-of-distribution data.



Caption: Left: CF-AUROC @ MSP $\geq$ 0.85 across methods and datasets (mit-b1). Right: T-Multi improvement over each GT baseline in percentage points. Error bars are \pm 1 std.

We see the spread between the best and worst detector widens with the type of shift, with roughly 2pp on ACDC, 7 on IDD and 11 on BDD. So, the choice of scoring method is slight for near-distribution cases such as ACDC and decisive under geographic shift. No post-hoc score is strong in all three datasets: as MaxLogit is strongest baseline comparison on BDD yet is the weakest by 2.7pp compared to GT-Dis on IDD. The right panel shows the fragility of the supervision target itself, as GT-Gap performs better over ACDC and IDD, and GT-Dis performs better on BDD. So, Δ_{GT} shows that neither of the two is reliably the runner-up. T-Multi is the strongest in all three datasets, and the teacher vs GT gap can be tested directly across seeds, where T-Multi beats the better of the two GT heads in every cell (\delta=+0.019; two-sided sign test p=0.004). Per-seed spread is small, so these orderings reflect the supervision signal rather than noise.

Paired Image Bootstrap **name can change**

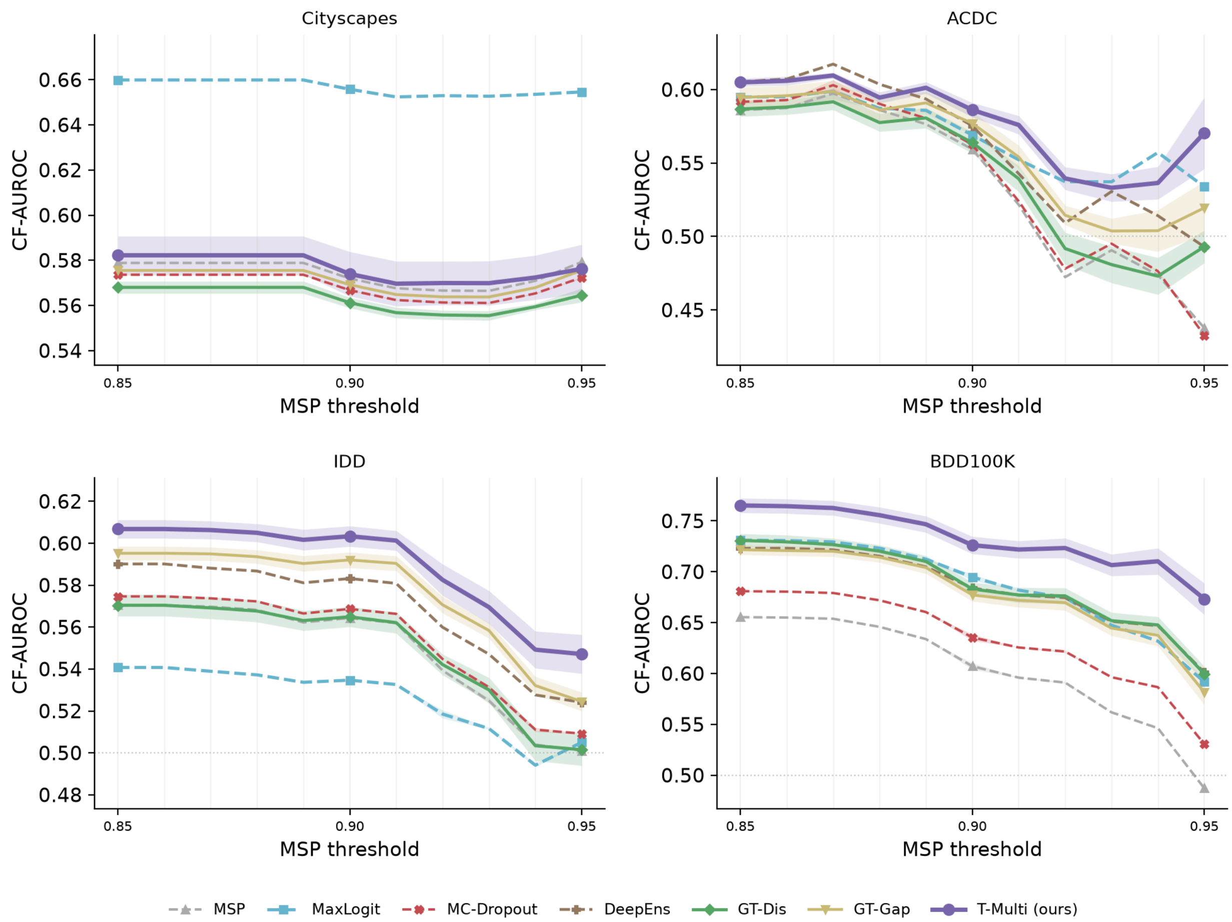
A complementary paired-image bootstrap (N=3000 resamples with replacement within each dataset with seed=42) attributes the T-Multi vs. GT-Dis advantage to BDD (Δ=+0.021, p=0.007) and IDD (Δ=+0.030, p<0.001) with no significance on ACDC (Δ=+0.014, p=0.23). Pooled, we find +0.022 (p=0.010) over GT-Dis and +0.020 (p=0.040) over GT-Gap. This pattern agrees with the within-condition analysis \ref{} to show the ACDC gains primarily come from condition-level separation rather than within-condition ranking.

High confidence failure detection

Increasing \tau does not simply make the task harder. Conditioning on MSP $\geq\tau$ removes the cases MSP as a confidence score would have removed. This leaves the failures MSP can not rank, which the guardrail is designed to target.

	τ	n	MSP	MaxL	Energy	MC-D	DeepEns	GT-Dis	GT-Gap	**T-Multi**
ACDC	0.85	397	.586	.595	.594	.592	**.605**	.587	.595	**.605**
	0.90	364	.559	.569	.568	.563	.575	.564	.577	**.586**
IDD	0.85	976	.570	.541	.539	.575	.590	.570	.595	**.607**
	0.90	959	.564	.535	.533	.569	.583	.565	.592	**.603**
	0.95	634	.501	.505	.505	.509	.524	.502	.524	**.547**
	0.97	231	.485	**.564**	**.564**	.498	.498	.488	.515	.471
BDD	0.85	987	.655	.731	.730	.681	.723	.731	.722	**.765**
	0.90	959	.607	.695	.694	.635	.684	.683	.677	**.726**
	0.95	622	.487	.592	.592	.531	.601	.600	.582	**.673**
	0.97	169	.341	.380	.380	.395	.471	.412	.453	**.566**

Caption: CF-AUROC at strict thresholds. Probabilistic post-hoc scores decay as τ rises compared to magnitude-based scores which are more resilient on IDD thought not on BDD. After $\tau=0.95$, multiple scores fall below chance, where MSP reaches 0.341, meaning that inside the subset of confident scores, a higher score is associated with higher risk, showing the silent failure case. This is not due to conditioning on MSP, as picking the BDD $\tau=0.97$ subset of $n{=}169$ and rather than ordering by MSP, ordering by MaxLogit, Energy, or Guardrail leaves MSP at 0.306, 0.323, and 0.263 respectively. So, self selection may even flatten MSP by ~4pp. Against BDD at $\tau=0.97$, T-Multi is the only method to score above chance, and its margin over the better GT head grows monotonitcally. IDD inverts at $\tau=0.97$, where T-Multi falls to 0.471 while MaxLogit and Energy reach 0.564, which holds across three seeds ($+13.8$, $+7.4$, $+6.5$,pp) and a paired-image bootstrap ($\Delta{=}{+}13.8$,pp at seed 42, $95%$ CI $[+4.7,+23.4]$, $p{=}0.003$). The cause is a degeneracy in the head rather than a strength of the baselines, as on IDD, the standard deviation of the guardrail score contracts $3.8\times$ between $\tau{=}0.85$ and 0.97 (.0139 to .0037) while the risk it ranks keeps the spread (.078 to .076). This leaves it four times flatter than the same head on BDD at the same threshold (.0145). This contraction tracks the training signal, as inside IDD's confident subset, the ratio of recoverable to misleading disagreement pixels falls to 1.75 compared to 2.68 on BDD. So, teh teacher-relative target the head was trained to predict is almost absent where it ranks. This is distinct from the night weakness \ref{}, where the risk is nearly constant and there is little to detect, compared to this case where the risk varies normally but the detector's predictions are more constant. We only consider subset sizes ≥ 150 samples (Hanley-McNeil standard error of 0.06), as reliability at the strict end is bounded by subset size rather than training variance.

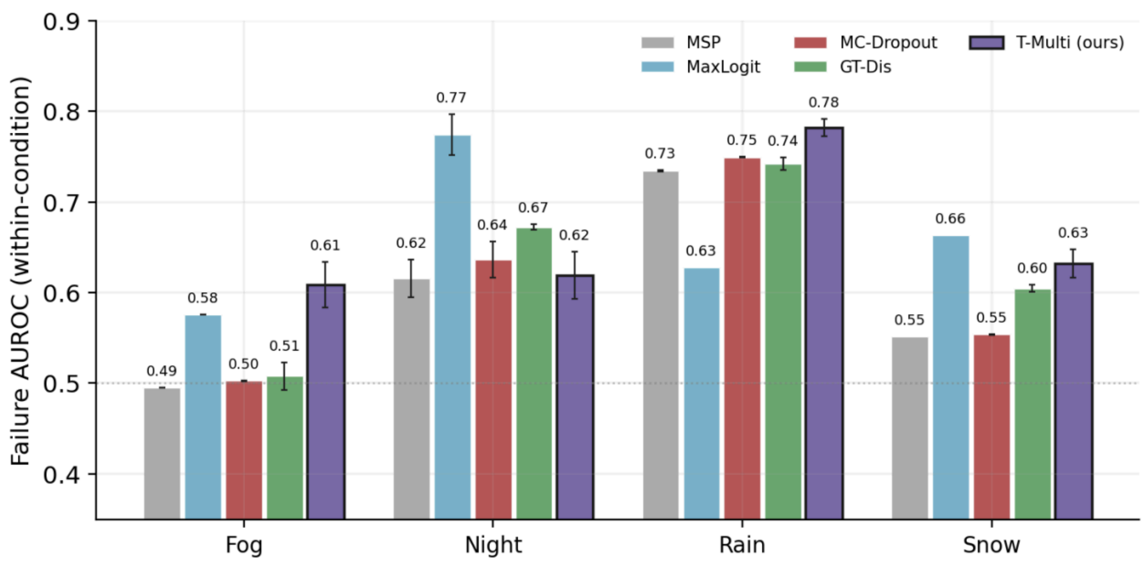


Caption: CF-AUROC swept at 0.01 steps (mit-b1). Lines are seed means with ± 1 std shaded. Threshold is plotted only at \geq 150 confident images, which truncates ACDC to $\tau=0.94$

We sweep \tau to confirm Table \ref is not a lucky sample, as T-Multi leads over all 11 BDD and IDD thresholds, while on ACDC we see no method consistently wins, through every gap between T-Multi and the best alternative is ≤ 2.2pp across all thresholds. Cityscapes shows the limitations, as the guardrail does not perform better in-domain, while MaxLogit clearly wins. In-domain, confident errors are visible in the student's logit magnitudes so a post-hoc score suffices and a learned head adds nothing. Under shift, the post-hoc methods struggle to track risk, and a teacher-supervised guardrail performs better as shift increases.

Within condition analysis

Cond.	MSP	MaxL	MC-D	GT-Dis	GT-Gap	**T-Multi**
Fog	.495	.576	.503	.519	.521	**.600**
Night	.607	**.760**	.625	.670	.659	.590
Rain	.734	.628	.749	.747	.762	**.764**
Snow	.552	**.664**	.554	.602	.628	.636
Macro-avg	.597	**.657**	.608	.634	.643	.648
Condition-only	0.932 (assigns condition mean risk, global cutoff)					
Pooled	.692	.927	.710	.910	.903	**.928**

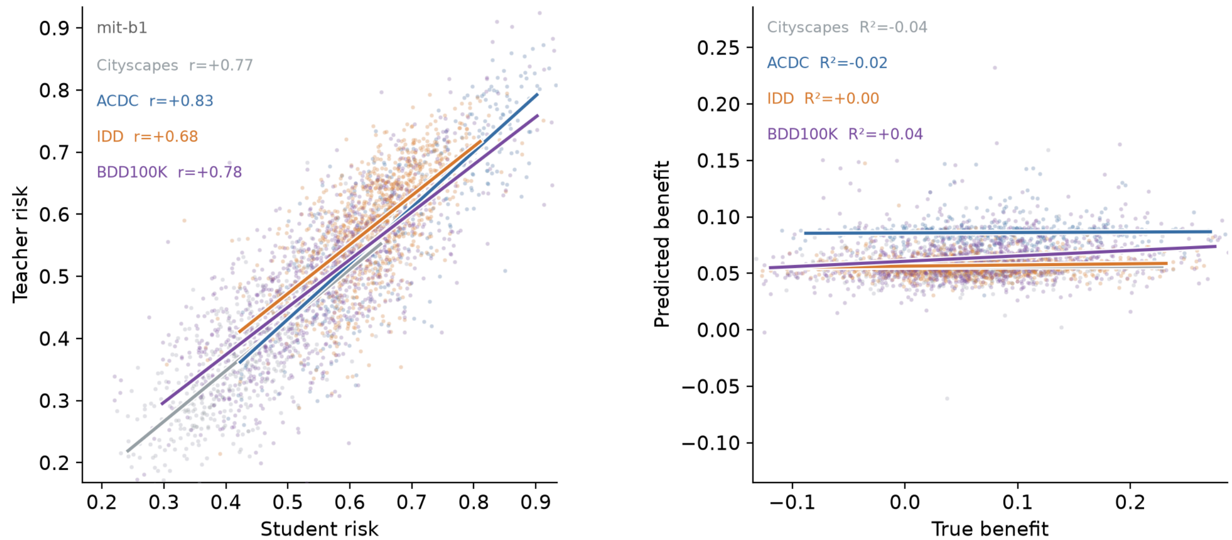


Supervision Mode Ablation

	MSP	Scalar	GT-Dis	GT-Gap	T-Dis	T-Gap	T-Multi
City	.579	.613	.566	.576	.574	.583	.571
ACDC	.587	.569	.589	.599	.582	.597	**.604**
IDD	.573	.570	.575	.592	.585	.617	.603
BDD	.657	.754	.738	.716	.745	.747	**.758**

Scalar baseline

The scalar did not work because of .



Caption: Student and teacher risks are tightly coupled, so the per-image benefit is a thin residual that cannot be recovered from the student's logits ($R^2 \le 0.04$ out-of-fold, right), motivating the dense per-pixel supervision.

Confidence features

Method	b0	b1	b2	Pooled
Energy score	.590	.595	.582	.589 ±.005
Guardrail + conf. features (ours)	**.624**	**.620**	**.589**	**.611 ±.017**
Δ vs. energy	**+3.4**	**+2.5**	**+0.7**	**+2.2**

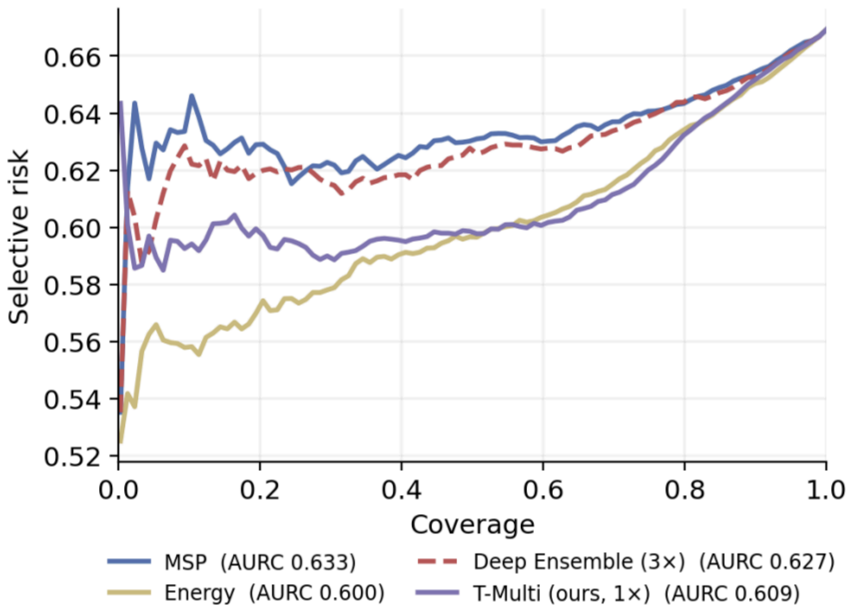
Caption:

Inference cost reliability comparison

Method	Latency (ms)	ACDC	IDD	BDD100K	Mean OOD
Student (MSP)	4.9	.437	.501	.487	.475
MaxLogit	4.9	.534	.505	.589	.543
MC-Dropout (4 passes)	25.2	.432	.509	.530	.490
Student + guardrail (T-Multi, ours)	16.8	.563	.541	.662	.589
Teacher (full inference)	26.1	—	—	—	—

- **Pareto Analysis**

- **Selective Deferral**



- **Selective Prediction**

	MSP	MaxL	Energy	TempMSP	MC-D	GT-Dis	GT-Gap	**T-Multi**
City	.422	**.415**	**.415**	.422	.422	.423	.422	.422
ACDC	.634	**.601**	.600	.625	.630	.610	.612	.609
IDD	.633	.636	.636	.635	**.632**	.634	**.632**	**.632**
BDD	.533	.529	.529	.531	.529	.521	.524	**.515**

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Discussion

Broader impacts

Conclusion